SURE A

A time to Shine

EXAMINATION
MOCK COUNCIL

SURE A EXAMINATION COUNCIL

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UCE CHEMISTRY 2026
MARKING GUIDE

SECTION A

*Respond to **both** items in this section.*

ITEM 1

Maria and her friend Gladys are students in senior one in a boarding school. They both have the same skills in washing clothes. On one weekend, they washed their uniforms to a fairly similar extent, with the same quantity of water from a borehole. Maria used cleaning agent **W** whereas Gladys used **X**. After washing, Maria realised that she had used more cleaning agent than what Gladys had used for the same number of uniforms. This left Maria wondering and as a result, she wants to know more about the cleaning agents.

Tasks:

(a) Categorise the cleaning agents **W** and **X**, giving a reason for your answer.

(b) Explain why Maria used more cleaning agent than Gladys.

(c) Explain the challenges associated with using cleaning agent **X** and their mitigations.

(d) Compare and contrast the products **X** and **W**.

RESPONSE

(a) Categorising the cleaning agents W and X

Cleaning agent **W** is a **soap**.

Cleaning agent **X** is a **soapless detergent** (a synthetic detergent).

Reason. Water from a borehole usually contains dissolved calcium ions, Ca^{2+}, and magnesium ions, Mg^{2+}, so it is hard water. In the same hard water and for the same number of uniforms, Maria's agent **W** was needed in a larger quantity than Gladys's agent **X**. This is the behaviour of a soap, which reacts with the calcium and magnesium ions and is used up before it can lather. A soapless detergent does not form scum with these ions and lathers readily in the same water.

In composition, a soap is the sodium salt of a long-chain carboxylic acid, for example sodium stearate, made from fats or oils. A soapless detergent is the sodium salt of a sulphonic acid or a sodium alkyl sulphate, made from petroleum products.

(b) Why Maria used more cleaning agent than Gladys

1. Borehole water contains dissolved calcium and magnesium salts, so it is hard water.
2. Soap is the sodium salt of a long-chain fatty acid, for example sodium stearate, $C_{17}H_{35}COO^{-}Na^{+}$. In hard water, the stearate ions react with the calcium and magnesium ions to form an insoluble precipitate called scum.

$$2C_{17}H_{35}COO^{-}(aq) + Ca^{2+}(aq) \rightarrow (C_{17}H_{35}COO)_2Ca(s)$$

$$2C_{17}H_{35}COO^{-}(aq) + Mg^{2+}(aq) \rightarrow (C_{17}H_{35}COO)_2Mg(s)$$

3. The soap that forms scum is used up and is no longer available for cleaning. Lather forms only after all the calcium and magnesium ions have been precipitated. Maria therefore had to add a large amount of **W** before the water could lather and clean the uniforms.

4. Gladys used a soapless detergent. The calcium and magnesium salts of soapless detergents are soluble, so no scum forms. All of the detergent remained available for cleaning, and a lather formed immediately. She therefore used less cleaning agent for the same number of uniforms.

(c) Challenges associated with using cleaning agent X and their mitigations

Challenge of using the soapless detergent X	**Mitigation**
Some soapless detergents, especially those with branched hydrocarbon chains, are not biodegradable. Bacteria cannot break them down, so they remain in water bodies.	Use soapless detergents with straight hydrocarbon chains, which are biodegradable, and use only the amount needed.
The persistent detergent produces a lot of foam on rivers and lakes. The foam reduces the amount of oxygen dissolving in the water and harms fish and other aquatic organisms.	Do not discharge wash water directly into streams, lakes or wetlands. Direct it to soak pits or treatment systems.
Many soapless detergents contain phosphates as additives. Phosphates enter water bodies and cause excessive growth of algae (eutrophication), which deprives aquatic organisms of oxygen when the algae decay.	Use phosphate-free detergents and treat waste water before releasing it into the environment.
Soapless detergents are made from petroleum products, which are non-renewable and make the detergent more expensive than soap.	Use measured amounts, avoid wastage, and use soap together with water softening where it is cheaper.
Strong detergents can irritate the skin and the eyes.	Wear gloves when washing, rinse hands thoroughly after use, and keep detergents out of the reach of children.

(d) Comparing and contrasting the products X and W

Similarities

1. Both are cleaning agents that remove grease and dirt from clothes and other surfaces.

2. Both have a hydrophilic (water-loving) head and a hydrophobic (water-hating) hydrocarbon tail.
3. Both lower the surface tension of water, and both remove grease by forming micelles that suspend the grease in water.
4. Both form a lather and clean well in soft water.

Differences

Feature	**W (soap)**	**X (soapless detergent)**
Raw materials	Made from animal fats or vegetable oils and an alkali such as sodium hydroxide (saponification).	Made from petroleum products (hydrocarbons) treated with concentrated sulphuric acid and then neutralised with sodium hydroxide.
Hydrophilic head	Carboxylate group, $-COO^-$	Sulphonate group, $-SO_3^-$, or sulphate group, $-OSO_3^-$
Action in hard water	Forms insoluble scum with Ca^{2+} and Mg^{2+} ions, so a lot is wasted.	Does not form scum because its calcium and magnesium salts are soluble, so little is needed.
Action in acidic water	Forms insoluble fatty acids and so cleans poorly.	Cleans well because it is the salt of a strong acid.
Biodegradability	Always biodegradable.	May be non-biodegradable if the chains are branched.
Source	Renewable raw materials	Non-renewable raw materials
Cost	Cheaper to produce	More expensive to produce

ITEM 2

John rears cattle which produce a lot of cow dung on his farm. Overtime, it has accumulated and attracts flies. He has been advised to make biogas from the cow dung using a biogas digester.

The main compound in the biogas used as fuel for cooking and lighting is methane. Methane burns according to the following equation.

$$CH_{4(g)} + 2\ O_{2(g)} \rightarrow CO_{2(g)} + 2\ H_2O_{(l)}$$

For effective use of methane, it should be measured in kilogrammes.

Before setting up the biogas digester, John would like to know about methane and the heat generated when 1 kg of methane is burnt.
You have been contacted by John for help.

Tasks:

(a) Classify the compound that John was advised to produce and give a reason for your answer.

(b) Explain the other properties of the compound to be produced by John.

(c) Determine the amount of heat produced when 1 kg of the compound is burnt during cooking.
(C = 12, H = 1, Heat of combustion of methane = ⁻885.4 kJ/mol)

(d) Suggest the risks of using the compound as a fuel and how they can be mitigated.

RESPONSE

(a) Classification of the compound

The compound John was advised to produce is **methane**, CH_4. Methane is a **covalent compound** (a simple molecular compound).

Reason. Methane is formed from two non-metals, carbon and hydrogen. Each carbon atom shares its four outer electrons with four hydrogen atoms to form four single covalent C–H bonds. Electrons are shared and not transferred, so no ions are formed and the compound exists as separate small molecules.

Further classification. Methane contains only carbon and hydrogen and has only single bonds, so it is also a hydrocarbon, and more specifically an **alkane** (a saturated hydrocarbon) with the general formula C_nH_{2n+2} where n = 1. The alkanes are a family within the covalent compounds, so both descriptions are correct.

(b) Other properties of methane

Physical properties (which follow from its covalent, molecular structure)

1. It is a colourless gas with no smell. A leak is therefore difficult to detect without an added odour.
2. It is less dense than air. Its relative molecular mass is 16 while the average relative molecular mass of air is about 29. Methane therefore rises and disperses upwards when it leaks into open air.
3. It is almost insoluble in water because its molecules are non-polar, and it dissolves in organic solvents.
4. It has very low melting and boiling points because it is a simple molecular substance whose molecules are held together by weak intermolecular forces.
5. It does not conduct electricity because it has no free ions or free electrons.

Chemical properties

1. It is flammable. It burns in plenty of air with a blue, non-luminous flame to produce carbon dioxide and water and to release a large amount of heat. This is why it is a good fuel.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

2. In a limited supply of air, it burns incompletely to form carbon monoxide, which is poisonous, or carbon (soot).

$$2CH_4(g) + 3O_2(g) \rightarrow 2CO(g) + 4H_2O(l)$$

3. A mixture of methane and air can explode when ignited.

4. Being saturated, it is unreactive towards acids, alkalis and common oxidising agents. It reacts with chlorine in sunlight by substitution.

$$CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$$

(c) Heat produced when 1 kg of methane is burnt

Relative molecular mass of methane:

$$M_r\ (CH_4) = 12 + (4 \times 1)$$

$$M_r\ (CH_4) = 12 + 4$$

$$M_r\ (CH_4) = 16$$

Molar mass of methane $= 16\ \text{g/mol}$.

Converting the mass to grams:

$$1\ \text{kg} = 1 \times 1000\ \text{g}$$

$$1\ \text{kg} = 1000\ \text{g}$$

Number of moles in 1 kg of methane:

$$\text{Moles} = \frac{\text{Mass}}{\text{Molar mass}}$$

$$\text{Moles} = \frac{1000}{16}$$

$$\text{Moles} = 62.5\ \text{mol}$$

The heat of combustion is $-885.4\ \text{kJ/mol}$, so 1 mole of methane releases 885.4 kJ of heat.

$$\text{Heat produced} = \text{Moles} \times \text{Heat of combustion per mole}$$

$$\text{Heat produced} = 62.5 \times 885.4$$

$$\text{Heat produced} = 55{,}337.5\ \text{kJ}$$

The enthalpy change is $\Delta H = -55{,}337.5$ kJ, where the negative sign shows that the heat is released (the reaction is exothermic).

The amount of heat produced when 1 kg of methane is burnt is about 55,338 kJ (55.3 MJ).

(d) Risks of using methane as a fuel and their mitigations

Risk	**Mitigation**
Methane is highly flammable. A leak near a flame or spark can cause a fire or an explosion when it mixes with air in the right proportion.	Keep the digester, pipes and stove away from naked flames and sparks. Install a pressure-relief valve and flame traps, and provide ventilation where the gas is stored and used.
Methane is colourless and odourless, so leaks are not noticed.	Check pipes, joints and valves regularly, test joints with soapy water, and add a strong-smelling substance to the gas so that leaks can be smelt.
Burning in a poor supply of air produces carbon monoxide, which is poisonous and can cause death by suffocation indoors.	Use burners that allow enough air to enter so that combustion is complete, cook in well-ventilated rooms, and keep the stove clean.
Biogas contains traces of hydrogen sulphide, which is toxic, corrosive and has an unpleasant smell.	Pass the gas through a scrubber, for example iron filings, before use and use corrosion-resistant fittings.
Methane and carbon dioxide are greenhouse gases. Leaked methane contributes to global warming, and burning produces carbon dioxide.	Prevent leaks by good construction and maintenance, and burn the gas instead of letting it escape.
Methane can collect in enclosed spaces and displace oxygen.	Never enter a digester pit without ventilation, and build the digester in an open area away from the house.

SECTION B

Part I

Respond to only **one** item in this part.
Responses should be written in the answer booklet(s) provided.

ITEM 3

Production of textiles and treatment of water are some of the industrial processes that use large amounts of sodium hydroxide as a raw material. Hence its continuous supply is important. The production of sodium hydroxide requires sodium chloride as a raw material.

An investor has proposed setting up a sodium hydroxide manufacturing factory in an area where sodium chloride is abundant. The community members wish to know about the production process, its socio-economic benefits and how any side effects can be mitigated.

You are required to prepare a presentation to the community members.

Task:
Write down your presentation.

RESPONSE

PRODUCTION OF SODIUM HYDROXIDE FROM SODIUM CHLORIDE

A presentation to the community members

1. Introduction

Good day, members of the community. Sodium hydroxide, NaOH, is also called caustic soda. It is a raw material for making textiles, soap and paper, and it is used for treating water. A factory that makes sodium hydroxide from the sodium chloride found in our area would supply these industries locally. Today I will explain how sodium hydroxide is made, the benefits the factory will bring to us, and how its side effects will be reduced.

2. Raw materials

1. Sodium chloride, NaCl, obtained as a concentrated solution called brine.
2. Water.
3. Electricity.

3. The production process: electrolysis of brine

Sodium hydroxide is made by the electrolysis of concentrated sodium chloride solution in an electrolytic cell. The steps are as follows.

Step 1: Preparing the brine. Concentrated brine is purified by adding sodium carbonate and sodium hydroxide to remove calcium and magnesium ions as insoluble precipitates, which are filtered off.

Step 2: Electrolysis. The purified brine is fed into a cell that has an inert titanium anode and a steel cathode. The anode and the cathode compartments are separated by a diaphragm or an ion-exchange membrane so that the products do not mix.

The brine contains the ions Na^+ and Cl^- from the sodium chloride and H^+ and OH^- from the water.

At the anode (positive electrode). Chloride ions are discharged in preference to hydroxide ions because the brine is concentrated. Chlorine gas is given off.

$$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$$

At the cathode (negative electrode). Hydrogen ions are discharged in preference to sodium ions because hydrogen is lower than sodium in the reactivity series. Hydrogen gas is given off.

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

In the solution. The Na^+ ions and OH^- ions remain in the solution around the cathode, which becomes a solution of sodium hydroxide.

Overall equation:

$$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$$

Step 3: Concentrating the product. The solution leaving the cell contains sodium hydroxide and unreacted sodium chloride. It is evaporated so that most of the sodium chloride crystallises out and is returned to the process, leaving a concentrated sodium hydroxide solution, which is stored or sold as pellets or solution.

Useful by-products: chlorine is used to treat water and to make bleach and plastics, and hydrogen is used as a fuel and to make ammonia and margarine.

4. Socio-economic benefits

1. Jobs will be created for the people of the area during construction and in the factory, for example as technicians, drivers, security staff and suppliers of services.
2. Sodium chloride found in the area will be put to use, which increases its value to the community.
3. Sodium hydroxide will be available to textile, soap and water-treatment industries without being imported, which saves foreign exchange and may lower prices.
4. By-products such as chlorine and hydrogen provide further products that can be sold.
5. The government and local authorities will collect taxes and levies that can be used for schools, health centres and roads.
6. Roads, electricity and water supply in the area will be improved, and local people will gain skills.
7. Small businesses such as food vendors, transport and housing will grow around the factory.

5. Side effects and how they will be mitigated

Side effect	**Mitigation**

Side effect	Mitigation
Chlorine gas is poisonous and can leak, harming the eyes and lungs.	Install gas detectors and alarms, keep the chlorine in sealed pipes and tanks, and absorb any escaping chlorine in sodium hydroxide solution to make bleach.
Waste water containing sodium hydroxide and brine raises the pH and salinity of streams and soil, which harms fish, crops and drinking water.	Neutralise waste water with an acid and treat it before release, recycle the brine, and never discharge waste into rivers or wetlands.
Mercury cells release mercury, which is highly toxic.	Use the diaphragm or membrane cell instead of the mercury cell.
Hydrogen is highly flammable and can explode.	Store and pipe it safely, keep sparks and flames away, and ventilate the plant.
Sodium hydroxide is corrosive and burns the skin and the eyes.	Provide protective clothing such as gloves and goggles, safety showers and training for workers.
The process uses a large amount of electricity, which may cause shortages or add to pollution if it comes from burning fuels.	Use hydroelectric or solar power, and use energy-efficient cells.
Noise, dust and heavy traffic, and loss of land used for farming.	Plant trees as a buffer, control noise and dust, compensate affected families fairly, and involve the community in monitoring the factory.

6. Conclusion

If the factory is built and managed with the safety and environmental measures described above, the community will gain jobs, income and better services, while the risks to people and the environment will be kept low. Thank you for listening. I welcome your questions.

ITEM 4

Climate change is a major challenge that has far reaching effects on humanity. One way of mitigating climate change is to reduce carbon emissions into the atmosphere. This can be done by reducing over dependence on carbon-based fuels for energy.

In an attempt to provide alternative energy source, the authorities are planning to extend electricity distribution to rural areas. However, this necessitates importing pure aluminium for making aluminium cables. Imported pure aluminium is expensive but the cost can be

greatly reduced if manufactured locally. Fortunately, some of the areas have aluminium deposits that can be extracted and purified.

You have been chosen to make a presentation on the possibility of setting up an industry that will extract and produce pure aluminium locally. Your presentation should include; the production process, socio-economic benefits, side effects of the industrial process and their mitigations.

Task:
Write down the presentation.

RESPONSE

LOCAL EXTRACTION AND PRODUCTION OF PURE ALUMINIUM

A presentation on setting up an aluminium industry

1. Introduction

Ladies and gentlemen, extending electricity to rural areas needs a lot of aluminium for cables. Aluminium is suitable for cables because it is a good conductor of electricity, has a low density so the cables are light, is ductile so it can be drawn into wires, and resists corrosion because of a protective layer of aluminium oxide. Importing it is costly. Since our area has aluminium deposits, I will explain how pure aluminium can be produced here, the benefits, the side effects and how they can be reduced.

2. Raw materials

1. The ore **bauxite**, which is impure aluminium oxide, $Al_2O_3 \cdot 2H_2O$. Its main impurity is iron(III) oxide, Fe_2O_3.
2. Sodium hydroxide solution, used to purify the ore.
3. Cryolite, Na_3AlF_6, a solvent for the pure aluminium oxide.
4. Graphite (carbon) electrodes.
5. A large supply of electricity.

3. The production process

The process has two stages: purification of the ore and electrolysis of the pure aluminium oxide.

Stage 1: Purification of bauxite.

1. The bauxite is crushed and dissolved in hot concentrated sodium hydroxide solution. Aluminium oxide is amphoteric, so it dissolves to form sodium aluminate solution, but iron(III) oxide is basic and does not dissolve.

$$Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2NaAl(OH)_4(aq)$$

2. The mixture is filtered to remove the insoluble impurities, which form a red residue called red mud.

3. The filtrate is cooled and seeded with crystals of aluminium hydroxide so that aluminium hydroxide precipitates.

$$NaAl(OH)_4(aq) \rightarrow Al(OH)_3(s) + NaOH(aq)$$

4. The aluminium hydroxide is filtered, washed and strongly heated to give pure aluminium oxide (alumina). The sodium hydroxide is recycled.

$$2Al(OH)_3(s) \rightarrow Al_2O_3(s) + 3H_2O(g)$$

Stage 2: Electrolysis of aluminium oxide.

1. The pure aluminium oxide has a very high melting point of about 2050 °C. It is therefore dissolved in molten cryolite at about 950 °C. This saves energy and improves the electrical conductivity of the electrolyte.

2. The electrolyte is placed in a steel tank lined with graphite. The graphite lining is the cathode (negative electrode), and graphite rods dipped into the molten mixture are the anodes (positive electrodes).

3. The molten mixture contains Al^{3+} ions and O^{2-} ions. A direct current is passed through it.

At the cathode. Aluminium ions gain electrons and molten aluminium forms. It is denser than the electrolyte, so it sinks to the bottom of the tank and is tapped off.

$$Al^{3+}(l) + 3e^- \rightarrow Al(l)$$

At the anode. Oxide ions lose electrons and oxygen gas forms.

$$2O^{2-}(l) \rightarrow O_2(g) + 4e^-$$

The hot oxygen reacts with the graphite anodes and burns them away as carbon dioxide, so the anodes have to be replaced regularly.

$$C(s) + O_2(g) \rightarrow CO_2(g)$$

Overall equation:

$$2Al_2O_3(l) \rightarrow 4Al(l) + 3O_2(g)$$

The molten aluminium is cast into ingots and then drawn into wires for cables.

4. Socio-economic benefits

1. The industry will create jobs in mining, transport, the plant and related services.

2. Locally made aluminium costs less than imported aluminium, which saves foreign exchange and makes electricity connections cheaper.

3. Cheap cables will speed up the extension of electricity to rural areas, which supports businesses such as milling, welding, hairdressing and cold storage, and improves schools and health centres.

4. Electricity reduces dependence on firewood, charcoal and other carbon-based fuels, which reduces carbon emissions and protects forests.

5. The country will earn income from selling aluminium for cables, roofing sheets, cooking utensils and other products, and the government will collect taxes.

6. Roads and services will improve, and people will gain new skills.

5. Side effects and their mitigations

Side effect	**Mitigation**
Red mud is strongly alkaline and contains iron oxide. It pollutes land and water and kills vegetation.	Store it in lined ponds, neutralise it before disposal, and reuse it, for example in making bricks.
The electrolysis uses a very large amount of electricity.	Use hydroelectric or solar power and recycle aluminium, since recycling uses much less energy than extraction.
Carbon dioxide is released when the graphite anodes burn, and fluoride gases from cryolite damage plants and harm health.	Fit scrubbers to capture the fluoride gases, use efficient cells, and plant trees around the factory.
Mining bauxite causes loss of vegetation, soil erosion, dust and land degradation.	Rehabilitate mined land by refilling and replanting, suppress dust with water, and mine only the areas that are approved.
Hot waste water and noise from the plant.	Cool the water before release and use noise barriers.
Local families may lose land.	Compensate affected people fairly and involve the community in decisions.

6. Conclusion

Producing pure aluminium locally is possible. The process of purifying bauxite and electrolysing the pure aluminium oxide in molten cryolite will provide cheaper cables for rural electrification, create jobs and reduce carbon emissions, provided the side effects are controlled using the measures above. Thank you.

Part II

*Respond to only **one** item in this part.*
Responses should be written in the answer booklet(s) provided.

ITEM 5

A city is experiencing many environment-related challenges, which affect the sustainable use of its natural resources.

In response to the challenges, the Environmental Management Authority (EMA) is creating awareness about sustainable use of resources for safe and clean environment under the theme: **My Environment, My Responsibility**. EMA has asked students to participate in the awareness by writing articles to be included in the EMA magazine. The article should include; category of resources, its composition, impact of human activity on the resources and benefits of the resources.

Task:
Write the article to be submitted to EMA.

RESPONSE

MY ENVIRONMENT, MY RESPONSIBILITY

Using our natural resources sustainably

Introduction

Everything we need to live comes from our environment. The air we breathe, the water we drink, the soil in which we grow food and the minerals and fuels we use for industry are all natural resources. Because our city is facing many environmental problems, every one of us must learn how these resources are made up, how our actions damage them and how we can use them wisely.

Category of resources

Natural resources are put into two main categories.

1. *Renewable resources* can be replaced naturally within a short time if they are used carefully. Examples are air, water, soil, forests and sunlight.
2. *Non-renewable resources* are used up faster than they form and cannot be replaced within a human lifetime. Examples are fossil fuels such as petroleum and coal, and minerals such as copper, iron ore and limestone.

This article focuses on the three resources that support our daily life in the city: air, water and soil.

Composition of the resources

Air. Air is a mixture of gases. Dry air is about 78% nitrogen, 21% oxygen, about 1% argon and other noble gases, and about 0.04% carbon dioxide, with varying amounts of water vapour.

Water. Pure water is a compound of hydrogen and oxygen, H_2O, in which two atoms of hydrogen are joined to one atom of oxygen. Water in nature also contains dissolved gases, for example oxygen, and dissolved salts of calcium, magnesium and sodium, together with suspended particles.

Soil. Soil is a mixture of mineral particles such as sand, silt and clay, humus (decayed plant and animal matter), water, air and living organisms such as bacteria, fungi and earthworms. It also contains mineral salts that plants need, for example nitrates, phosphates and potassium salts.

Impact of human activity on the resources

1. *On air.* Burning fuels in vehicles, factories, kitchens and open rubbish fires releases carbon dioxide, which adds to global warming, carbon monoxide, which is poisonous, sulphur dioxide and oxides of nitrogen, which cause acid rain, and smoke and dust, which cause breathing problems.

2. *On water.* Dumping solid waste, plastic and chemicals into drains, rivers and lakes pollutes the water. Untreated sewage and industrial effluent spread disease and kill aquatic life. Fertilisers and detergents washed into water bodies cause algae to grow excessively, which uses up the oxygen in the water (eutrophication). Draining wetlands reduces the natural cleaning and storage of water.

3. *On soil.* Cutting down trees and poor farming expose soil to erosion. Overuse of chemical fertilisers and pesticides and dumping of wastes such as plastics, batteries and oil contaminate the soil and reduce its fertility.

Benefits of the resources

1. Air provides oxygen for breathing and burning, nitrogen for making fertilisers, and carbon dioxide for photosynthesis in plants.

2. Water is needed for drinking, cooking, washing, farming, generating hydroelectric power and industry, and it supports aquatic life.

3. Soil supports crops, trees and grazing, and so provides food, timber and income.

4. Minerals and fuels provide energy and raw materials for building, manufacturing and transport.

What we must do

Protect trees and wetlands, dispose of waste properly and recycle plastics, use clean energy such as biogas and solar power, treat waste water before releasing it, and practise soil

conservation such as planting trees and terracing. Our environment is our home, and looking after it is our responsibility.

By a student, for the EMA magazine.

ITEM 6

Most urban dwellers in Uganda depend on charcoal as a fuel for cooking. This has negatively affected the forest cover in the country. In one of the sub counties, National Forest Authority (NFA) has organised a sensitisation workshop. The aim of the workshop is to enlighten the public about conservation of forests, dangers of charcoal burning, social benefits of the forests and alternative sources of fuel.

You have been chosen to represent the youth of the area to demonstrate their level of awareness.

Task:
Compose the sensitisation message.

RESPONSE

SAVE OUR FORESTS: SAY NO TO CHARCOAL BURNING

A sensitisation message from the youth of the sub county

Distinguished guests from the National Forest Authority, elders, parents and fellow youth, good day.

Why we must conserve our forests

Forests are the lungs of our country. Trees take in carbon dioxide and give out oxygen during photosynthesis, which keeps the air clean and slows climate change. Their roots hold the soil together, so they prevent soil erosion and landslides and protect our rivers and wetlands. They also bring rain and keep our area cool. When we cut down trees faster than they can grow, we lose all of these gifts.

The dangers of charcoal burning

Charcoal is made by heating wood with very little air in a kiln. This has serious effects.

1. It destroys forests because many trees are cut down to make a small amount of charcoal, which leads to bare land, soil erosion, loss of rainfall and drying of streams.
2. Burning wood in the kiln and burning charcoal in the stove release carbon dioxide, which adds to global warming. When the air supply is poor, a poisonous gas called carbon monoxide is produced.
3. Carbon monoxide has no colour and no smell. People who cook with charcoal in closed rooms can be poisoned and may even die.

4. The smoke and soot irritate the eyes and the lungs and cause coughs and other breathing problems, especially among women and children who spend a lot of time in the kitchen.

5. Kiln fires can spread and burn farms, forests and homes.

Social benefits of forests

Forests give us firewood from planted trees, timber for building and furniture, poles, fruits, honey, mushrooms and herbs for medicine. They provide jobs and income through activities such as bee keeping, tree nurseries and tourism. They are also homes for wild animals and birds, and they provide places for recreation and for cultural and spiritual practices.

Alternative sources of fuel

1. **Biogas** made from cow dung, kitchen waste and other organic matter. It burns cleanly and the left-over slurry is a good fertiliser.

2. **Liquefied petroleum gas (LPG)**, which burns cleanly and quickly.

3. **Briquettes** made from waste such as sawdust, crop residues, waste paper and charcoal dust, which reduces the need to cut trees.

4. **Energy-saving stoves**, which use much less charcoal or firewood to cook the same food.

5. **Electricity and solar cookers**, which produce no smoke.

6. **Planted woodlots** for firewood, which are used in a way that new trees are planted to replace those that are cut.

Our call to action

We, the youth, call upon everyone to plant at least a tree for every tree cut, to stop cutting trees for charcoal, to switch to cleaner fuels, to cook in well-ventilated kitchens, and to report illegal tree cutting and charcoal burning to the authorities. Let us protect our forests today so that they can protect us tomorrow.

Thank you very much.